Technology Stack

Date	14 July 2026
Team ID	
Project Name	Automated Credit Card Application Screening using ML
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, Flask (Jinja2)	Provides a clean, user-friendly interface for bank analysts and customers to input financial profiles and view real-time approval predictions via a responsive dashboard.
2	Backend / Server-Side	Python, Flask, Scikit-learn	Implements the business logic and processing. Loads four optimized classification models (Logistic Regression, Random Forest, XGBoost, Decision Tree) to generate real-time screening results.
3	Machine Learning Pipeline	Scikit-learn, XGBoost, NumPy, Pandas	Handles feature engineering, multi-class to binary label conversion (high/low risk), and data normalization using StandardScaler to ensure optimal model convergence.
4	Cloud / Deployment	IBM Watson Machine Learning	Deploys the best-performing model (determined via Grid Search cross-

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			validation) as a scalable API endpoint for real-time predictions on demand.
5	Explainer & Diagnostics	Matplotlib, SHAP / Lime	Generates feature importance plots and explainability visualizations to help compliance officers understand the key factors driving approval/ rejection decisions.
6	Third-Party APIs / Other Tools	Git, GitHub, Joblib	Facilitates version control, experiment tracking, and efficient serialization (saving) of tuned models for quick loading within the Flask environment.